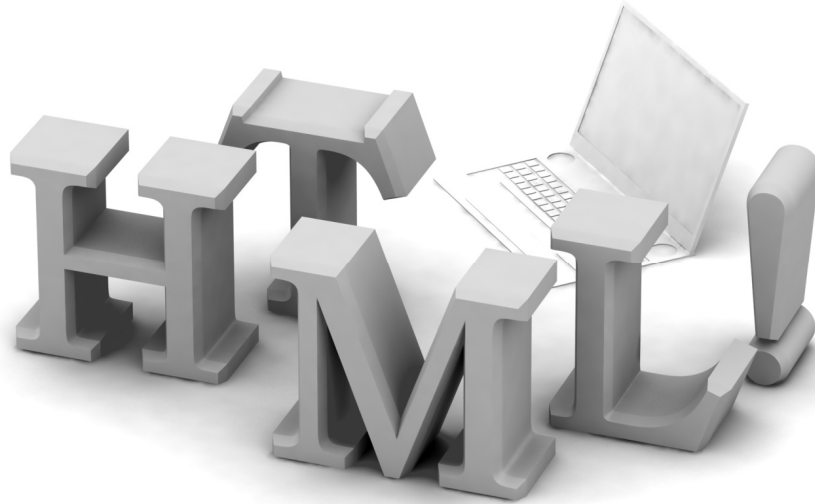


HTML Language



Origins of HTML...

HTML = Hyper Text Markup Language

Simple markup language that describes the content of web pages (without requiring programming skills)

HTML derives from SGML (Standard Generalized Markup Language)

A meta-language to create languages

An HTML document is a **text file** that contains the HTML code describing the page

→ A web page can be created using any **text editor**

The HTML language

... origins of HTML

HTML was created with the aim to be *device independent*

Its original purpose was to describe the *semantic structure* of a scientific document,
not its appearance (task left to the browser)

But the WWW spread very rapidly...

From scientific and military communities to many other fields (especially commercial contexts)

The language was thus progressively modified to include formatting information

Against its initial aim

Tags

Define the structural elements of an HTML document

Specify the semantic meaning of a portion of the document, connections to other documents, images, or other multimedia elements to be included, etc.

Are enclosed within the ‘<’ and ‘>’ characters

e.g., <body>, <h1>, ...

In most cases, are used “in pair”

each “opening tag” has a corresponding “closing tag”
(e.g., <body> ... </body>, <head> ... </head>, ...)

HTML versions

- HTML 5.3 is the latest “standardized” (*W3C official recommendation*) version of HTML (14 December 2017)
- Now HTML5 is a *living standard*: <https://html.spec.whatwg.org/>
The WHATWG (Web Hypertext Application Technology Working Group) was founded in 2004 by individuals from Apple Inc., the Mozilla Foundation and Opera Software (now it also includes Google)
- Previous version (before HTML 5): 4.01

The HTML language

HTML Living Standard

The screenshot shows the HTML Living Standard website. At the top, the word "HTML" is in large green letters, followed by "Living Standard — Last Updated 23 October 2025" in smaller green text. Below this is a grid of 12 buttons with various links and descriptions. At the bottom left, there is a "Table of contents" section with a list of five items.

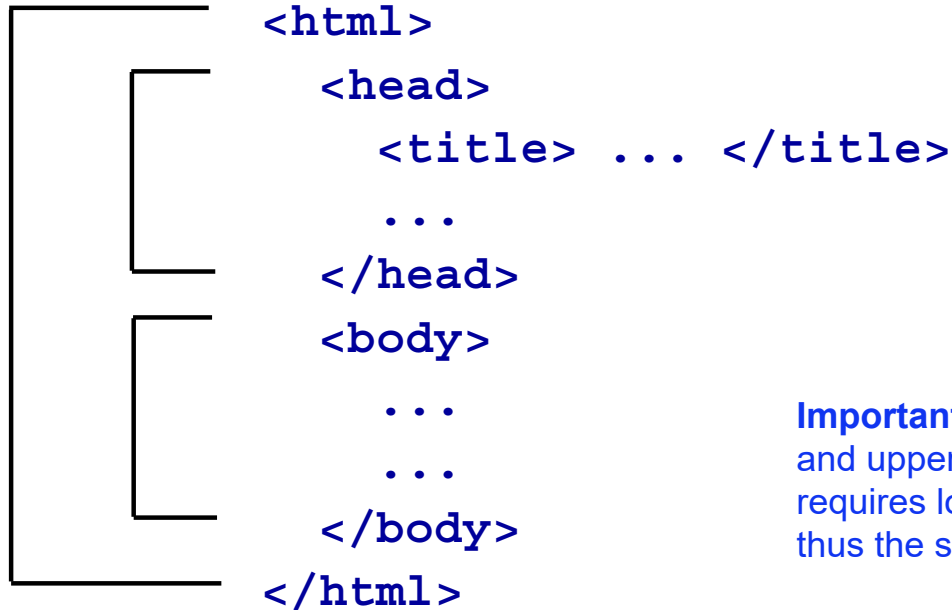
HTML
Living Standard — Last Updated 23 October 2025

One-Page Version html.spec.whatwg.org	Multipage Version /multipage	Version for Web Devs /dev	PDF Version /print.pdf	Translations 日本語・简体中文	FAQ on GitHub	Chat on Matrix	
Contribute on GitHub whatwg/html repository	Commits on GitHub	Snapshot as of this commit	Twitter Updates @htmlstandard	Open Issues filed on GitHub	Open an Issue whatwg.org/newbug	Tests web-platform-tests html/	Issues for Tests ongoing work

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- 4 The elements of HTML
- 5 Microdata

General structure of an HTML document



Important: tags can be both lowercase and uppercase. However, XHTML requires lowercase tags... lowercase is thus the suggested choice

Basic tags

<html>

Specifies that the file contains HTML code

<head>

Identifies the “beginning” section of the document

<title>

Specifies a title for the document

<body>

Defines the main section of the document

Headings and paragraphs

Six *heading* levels

Have a semantic (or logical) meaning

(<h1> → most important, <h6> → least important)

Paragraphs define “ordinary text”

The way they are displayed is decided by the browser (like for all semantic tags)

Spaces and line breaks are ignored

HTML was designed to be **device-independent**

Attributes

Tags can have *attributes*

For example, to add a title to a paragraph:

```
<p title="Important content">Some text</p>
```

Attributes define properties for tags

- A tag can have more than one attribute
- Structure: `attributename="attributevalue"`

Deprecated attributes should never be used

Lists

Three possible kinds of lists:

- *Unnumbered lists* → tags `<ul>` and `<li>`
To create non-ordered lists
- *Numbered lists* → tags `<ol>` and `<li>`
To create ordered lists
- *Definition lists* → tags `<dl>`, `<dt>` and `<dd>`
To create a list of items, with a description of each one of them

Nested lists are also allowed

Lists within other lists' items

Preformatted text

Allows spaces and line breaks to be considered by the browser

The `<pre>` tag is the "exception that proves the rule"

Often used to include HTML or programming code in web pages

Otherwise, the code would be unreadable

Line breaks and horizontal rulers

Although HTML, by default, ignores line breaks, it is possible to force them

By means of the `<br>` tag (no closing tag is required)

Horizontal rulers are useful to separate different sections of the same document

By means of the `<hr>` tag, possibly with the `size` and `width` attributes

Example:

```
<hr size="4" width="50%">
```

Logical vs. physical styles

Logical styles

Tags or attributes that specify the **kind** (i.e., the semantic meaning) of content (while the way the content is displayed is decided by the browser)

Examples: `<strong>`, `<em>`, `<a>`, `title`, `href`, `type`, ...

Physical styles

Tags or attributes that specify the appearance of the content (i.e., the way the browser will display it)

Examples: `<font>`, `<center>`, `color`, `size`, `align`, ...

Escape sequences

Are used to include “special characters” in web pages

Such as non-English alphabet characters or characters that have a special meaning for HTML

Examples:

`<` = <

`>` = >

`&` = &

``` = è

`´` = é

`È` = È

`©` = ©

Links...

Allow to connect pages to other pages or files

Structure:

```
<a href="address">...</a>
```

Example:

```
<a href="https://web.unipv.it/">  
  University of Pavia  
</a>
```

Pathnames can be both *absolute* and *relative*

Examples:

```
<a href="/xxx/myfile.html">University of Pavia</a>  
<a href="xxx/myfile.html">University of Pavia</a>
```

Absolute paths refer to the *root* directory of the *web server*

... links

It is also possible to create links to sections within the same document or to specific parts of other documents

Example:

`<a href="#sect1">Section 1</a>` creates a link to “sect1” within the current document

`<a href="MyFile.html#sect1">Section 1</a>` creates a link to “sect1” in file `MyFile.html`

“sect1” will be defined by:

`<a name="sect1">This is Section 1</a>`

Links to e-mail addresses

Allow to automatically open the (possible) e-mail client with a click, to write the message

Example:

```
<p>  
  <a href="mailto:marco.porta@unipv.it">Marco Porta</a> is...  
</p>
```

Links to multimedia elements

A link can also create a connection to a multimedia file (image, animation, etc.)

The file path will be the value of the `href` attribute

Example:

```
<a href="MyImage.gif">Image 1</a>
<a href="MyMovie.mov">Movie 1</a>
```

A multimedia element can also be a link

Example:

```
<a href="MyImage.gif">
  
</a>
```

Inclusion of images

Tag `<img>` (does not need a closing tag)

Example:

```

```

It is possible to specify the size of the image

Although it is not strictly necessary...

Example:

```

```

“Alternative text” can be useful/necessary

Example:

```

```

Background color

Warning: deprecated attribute. **Do not use it!**

It is possible to specify a background color for the page

Attribute `bgcolor` of the `<body>` tag

Example:

```
<body bgcolor="yellow">
```

There are 16 predefined colors (*aqua, black, blue, fuchsia, gray, green, lime, maroon, navy, olive, purple, red, silver, teal, white, yellow*)

Hexadecimal codes can be used as well

Example: `bgcolor="#02fc16"`

Background image

Warning: deprecated attribute. **Do not use it!**

It is possible to specify a background image for the page

Attribute `background` of the `<body>` tag

Example:

```
<body background="MyImage.gif">
```

The image is repeated both horizontally and vertically

So that, regardless of its size, all the background is filled

Fonts

Warning: deprecated tag. **Do not use it!**

It is possible to specify the font (and its features) to be used for a portion of text

Family, color, size

Tag `<font>`

With attributes `face`, `color` and `size`

Example:

```
<h1>  
  <font face="sans-serif" color="red" size="+2">  
    This a header with font <em>sans-serif</em>, red  
    color and size <em>+2</em>  
  </font>  
</h1>
```

Tables...

Allow tabular arrangements of data

Useful to structure intrinsically tabular information

In the past, they were also used to create page layouts

Now the layout **must** be defined using Cascading Style Sheets!

Are described in terms of *rows* and *columns*

- Table → tag `<table>`
- Rows → tag `<tr>`
- Columns → tag `<td>` and `<th>`

... tables...

Table with n rows and m columns:

<pre><table></pre>			
<pre> <tr></pre>			
<pre> <td>Row 1, Column 1</td></pre>			
<pre> </pre>			
<pre> <td>Row 1, Column m</td></pre>			
<pre> </tr></pre>			
<pre> </pre>			
<pre> <tr></pre>			
<pre> <td>Row n, Column 1</td></pre>			
<pre> </pre>			
<pre> <td>Row n, Column m</td></pre>			
<pre> </tr></pre>			
<pre></table></pre>			

Cells of row 1

Row 1

Table

Cells of row n

Row n

... tables

Warning: some of these attributes are deprecated in HTML5

Attributes of tag <table>

width → table width; **border** → border size; **align** → horizontal alignment;
valign → vertical alignment; **cellpadding** → space between cell content and cell border;
cellspacing → space between cells; **bgcolor** → background color;
background → background image

Attributes of tag <tr>

align → horizontal alignment; **valign** → vertical alignment;
bgcolor → background color

Attributes of tags <td> and <th>

width → cell width; **height** → cell height; **bgcolor** → background color; **background** → background image; **align** → horizontal alignment;
valign → vertical alignment; **colspan** → number of columns occupied by the cell horizontally; **rowspan** → number of rows occupied by the cell vertically

Forms...

Allow the user to provide inputs to web pages

Example:

Registration form

Name

Surname

Password

Gender ☐ M ☐ F

I heard about this service through ☐ Internet ☐ Press ☐ Television

I wish to receive a message every

Comments

Upload resumé

... forms...

Forms are created through the `<form>` tag

The `<input>` tag allows to define several *controls* (input elements)

The `type` attribute specifies the kind of control

- `text` → text input
- `password` → password input
- `checkbox` → yes/no option
- `radiobutton` → mutually exclusive option
- `button` → button
- `submit` → button “to send” data
- `reset` → button to reset the form fields
- `file` → file upload
- `hidden` → to send “hidden” data

Other tags/controls

- `<textarea>` → multiple-row text input
- `<select>` and `<option>` → drop-down menu
- `<label>` → label

HTML5 introduces new types for controls

Form data must be sent somewhere...

- Typically, to a server which will process them in some way (`action` attribute of the `<form>` tag)
- Two possible “methods”:
GET and POST (`method` attribute of the `<form>` tag)

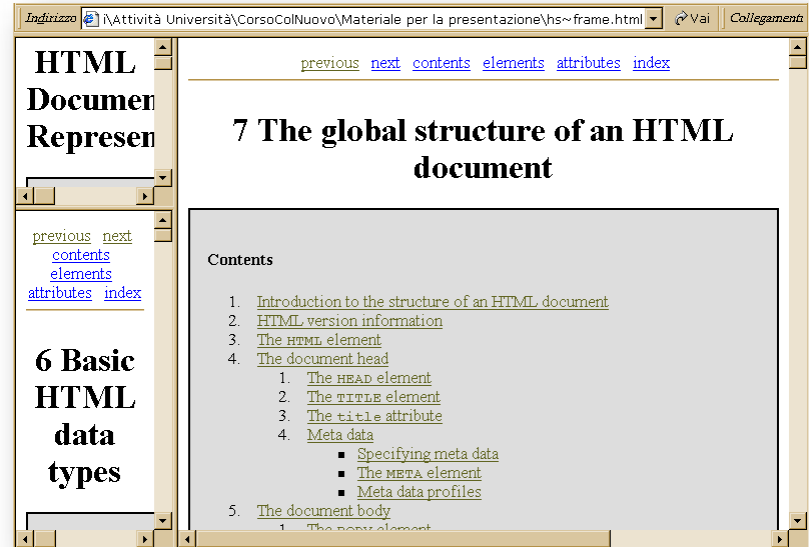
Frames...

Warning: very old and deprecated design approach. **Never use frames!**

Subdivide the browser window into “sub-windows”

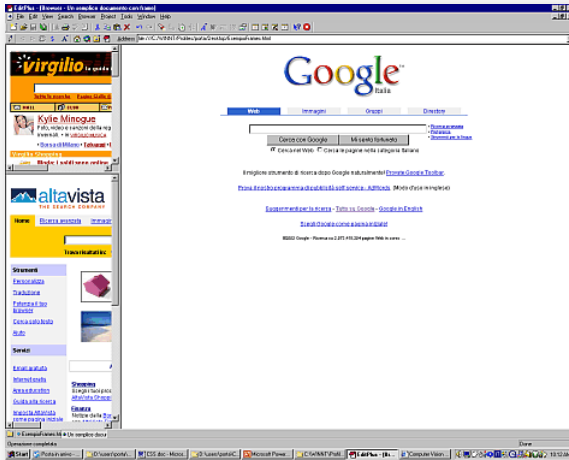
Example:

Frames must be avoided because of accessibility and usability reasons



The HTML language

... frames



```
<html>
  <head>
    <title>A simple frame document</title>
  </head>
  <frameset cols="20%, 80%">
    <frameset rows="100, 200">
      <frame src="http://www.virgilio.it/">
      <frame src="http://www.altavista.com/">
    </frameset>
    <frame src="http://www.google.com/">
  </frameset>
  <noframes>
    <p>Sorry, this website uses frames...</p>
  </noframes>
</frameset>
</html>
```